

A New Perspective to Boost Performance: Fairness for Medical Federated Learning

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Motivation & Problem

Motivation:

- ▶ Improve fairness in FL for sustainable hospital collaboration.
- ▶ Crucial in medical applications with diverse datasets.

Problem:

- ▶ Existing fair FL methods **ignore domain shift**.
- ▶ Performance disparities reduce incentives for hospital participation.

Challenge:

- ▶ How to ensure fairness while accounting for layer-wise feature shifts across hospitals?

Federated Learning in Healthcare

Federated Learning (FL):

- ▶ Enables collaboration across hospitals without sharing raw data.
- ▶ Leverages diverse datasets for a stronger global model while preserving privacy.

Why Fairness Matters:

- ▶ Standard FL emphasizes only **global performance**.
- ▶ Risk: some hospitals benefit less (or even suffer), reducing motivation to participate.
- ▶ Fairness ensures all hospitals gain, sustaining collaboration.

Limitations of Existing Fair FL Methods

Current focus:

- ▶ Collaboration fairness (resource allocation).
- ▶ Performance fairness (client-level outcomes).

Key limitation:

- ▶ Assign a **single weight per client**, ignoring **layer-wise feature shifts**.
- ▶ Domain shifts cause layer-level representation differences → suboptimal aggregation.

Challenge:

- ▶ Need aggregation that accounts for **layer-wise representation differences**.

Proposed Solution: Fed-LWR

Fed-LWR: Layer-Wise Re-weighting

- ▶ Uses the **global model as anchor** and compares local models to it via **layer-wise CKA similarity**, capturing feature representation differences across hospitals.
- ▶ Translates these similarity scores into **layer-specific weights**, giving more influence to hospitals whose local features differ significantly from the anchor.
- ▶ Performs **layer-wise re-aggregation** using these weights, ensuring the final global model adapts fairly to diverse client data distributions.

Outcome:

- ▶ Better fairness/accuracy trade-off than state-of-the-art.
- ▶ Hospitals with weaker performance are not left behind.

Preliminaries: Federated Learning

Setup:

- ▶ K hospitals participate in FL via a trusted central server.
- ▶ Hospital $k \in [K]$ has private dataset D_k with n_k samples $\{(X_i, Y_i)\}_{i=1}^{n_k}$.
Each hospital trains a neural network $f = h_M \circ \dots \circ h_2 \circ h_1$ with M layers.

Federated Learning Objective:

- ▶ Minimize local empirical risk L_k for each hospital.
- ▶ Update global model via averaging:

$$w_t^G = \frac{1}{K} \sum_{k=1}^K w_t^k$$

where w_t^G and w_t^k are global and local model parameters at round t .

Performance Fairness

Definition: Model $\hat{w}$ is more **fair** than w if its performance is more uniform across the K hospitals.

Importance:

- ▶ Prevents disproportionate advantage/disadvantage to any hospital.
- ▶ Maintains incentives for all hospitals to participate.

Intuition:

- ▶ High average accuracy alone is not enough.
- ▶ Fairness considers **per-hospital performance balance**.
- ▶ The goal is to balance between **high** and **fair** performance.

Dice Score: Evaluation Metric

Definition: Measures overlap between predicted and ground truth segmentation.

$$\text{Dice} = \frac{2|X \cap Y|}{|X| + |Y|}$$

Range: $0 \rightarrow$ no overlap, $1 \rightarrow$ perfect overlap.

Interpretation:

- ▶ Sensitive to false positives/negatives.
- ▶ Crucial for medical image segmentation (MRI, retinal, etc.)

Advantages: Intuitive, region-focused, widely used.

Use in FL: Evaluates both accuracy and fairness across hospitals.

Hilbert-Schmidt Independence Criterion (HSIC)

Purpose: Measures dependence between two feature sets.

Given: Feature matrices from two layers:

$$U \in \mathbb{R}^{n \times p}, \quad V \in \mathbb{R}^{n \times q}$$

Gram Matrices:

$$K = UU^\top, \quad L = VV^\top$$

Centered Gram Matrices:

$$K' = HKH, \quad L' = HLH, \quad H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$$

HSIC Calculation:

$$\text{HSIC}(K, L) = \frac{1}{(n-1)^2} \text{tr}(K'L')$$

Interpretation: Higher HSIC $\rightarrow$ more dependent / similar feature representations.

Centered Kernel Alignment (CKA)

Purpose: Quantifies similarity between two layers' representations using HSIC.

CKA Formula:

$$\text{CKA}(U, V) = \frac{\text{HSIC}(K, L)}{\sqrt{\text{HSIC}(K, K) \cdot \text{HSIC}(L, L)}}$$

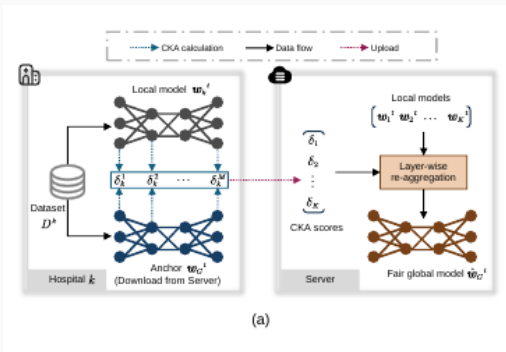
Interpretation:

- ▶ $\delta = 1$: Layers have identical representations
- ▶ $\delta = 0$: Layers are completely dissimilar

Use in Fed-LWR:

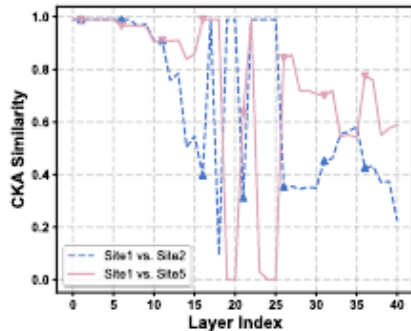
- ▶ Compute layer-wise similarity between local and global models.
- ▶ Lower CKA $\rightarrow$ higher weight for that hospital in re-aggregation.

Fed-LWR: Illustration and CKA Variation



(a) Overview of Fed-LWR: Layer-wise CKA similarity δ_k is computed between local and anchor models and used to re-aggregate for a fair global model $\hat{w}_t^G$.

Fed-LWR



(b) CKA variation across layers for two randomly selected client pairs, showing that model differences vary by layer.

Algorithm 1: Fed-LWR

Algorithm 1: Fed-LWR

Input: K hospitals, communication rounds T , learning rate η ,

Output: $\hat{\mathbf{w}}_G^T$

```
1 Initialize  $\mathbf{w}_G^0$  //  $\hat{\mathbf{w}}_G^0 = \mathbf{w}_G^0$ 
2 for round  $t = 1, 2, \dots, T$  do
3   for hospital  $k = 1, 2, \dots, K$  parallelly do
4      $\mathbf{w}_k^t \leftarrow \hat{\mathbf{w}}_G^{t-1}$ 
5      $\mathbf{w}_k^t \leftarrow \mathbf{w}_k^t - \eta \nabla \mathcal{L}_k$  // Local training
6   end
7    $\mathbf{w}_G^t \leftarrow \frac{1}{K} \sum_{k=1}^K \mathbf{w}_k^t$  // Averaging aggregation for the anchor
8   for hospital  $k = 1, 2, \dots, K$  parallelly do
9      $\delta_k \leftarrow [\delta_k^1, \dots, \delta_k^m, \dots, \delta_k^M]$ ,  $\delta_k^m \leftarrow \text{CKA}(\mathbf{U}_k^m, \mathbf{V}_k^m)$  // Eq. (3)
10  end
11   $\rho_k \leftarrow [\rho_k^1, \dots, \rho_k^m, \dots, \rho_k^M]$ ,  $\rho_k^m \leftarrow \frac{1 - \delta_k^m}{\sum_{i=1}^K (1 - \delta_i^m)}$  // Eq. (6)
12   $\hat{\mathbf{w}}_G^t \leftarrow \{\hat{\mathbf{w}}_{G,m}^t\}_{m=1}^M$ ,  $\hat{\mathbf{w}}_{G,m}^t \leftarrow \sum_{k=1}^K \rho_k^m \mathbf{w}_{k,m}^t$  // Eq. (3)
13 end
14 return  $\hat{\mathbf{w}}_G^T$ 
```

Fed-LWR: Representation Difference Estimation (Idea)

Goal: Quantify feature differences between local and global models for each hospital.

Motivation:

- ▶ Clients have heterogeneous data distributions → feature shift.
- ▶ **CKA similarity** can capture representation differences across models.
- ▶ Lower similarity → global model further from local optimum → poorer performance.

Fed-LWR: Representation Difference Estimation (Steps)

After local training in round t , the global model w_t^G (anchor) is sent to each hospital.

For hospital k , compute **CKA similarity** for each layer m :

$$\delta_k^m = \text{CKA}(U_k^m, V_k^m)$$

Construct the layer-wise similarity vector:

$$\delta_k = [\delta_k^1, \delta_k^2, \dots, \delta_k^M]$$

Fed-LWR: Representation Difference Estimation (Feature Matrices)

Feature matrices used in CKA:

$$U_k^m = h_m \circ h_{m-1} \circ \dots \circ h_1([X_1, \dots, X_{n_k}], w_t^{k,m} \cup \dots \cup w_t^{k,1})$$

$$V_k^m = h_m \circ h_{m-1} \circ \dots \circ h_1([X_1, \dots, X_{n_k}], w_t^{G,m} \cup \dots \cup w_t^{G,1})$$

U_k^m : feature matrix from local model k at layer m .

V_k^m : feature matrix from global anchor at layer m .

Fed-LWR: Representation Difference Estimation (Interpretation)

- ▶ Higher $\delta_k^m \rightarrow$ local and global representations are similar.
- ▶ Lower $\delta_k^m \rightarrow$ greater feature difference, meaning:
 - ▶ Global model is further from local optimum.
 - ▶ Poorer performance expected for hospital k .
- ▶ These similarity scores δ_k^m will later guide **layer-wise weighting in aggregation**.

Fed-LWR: Aggregation Weights Computation

Goal: Assign fair weights to hospitals based on feature differences.

Step 1: Convert CKA similarity to weights for each layer m :

$$\rho_k^m = \frac{1 - \delta_k^m}{\sum_{i=1}^K (1 - \delta_i^m)}$$

Interpretation:

- ▶ Hospitals with larger feature differences get higher weights.
- ▶ Ensures clients with poorer performance influence the global model more fairly.

Fed-LWR: Layer-wise Aggregation & Outcome

Step 2: Re-aggregate each layer of the global model:

$$\hat{w}_t^{G,m} = \sum_{k=1}^K \rho_k^m w_t^{k,m}$$

Step 3: Combine layers to obtain fair global model:

$$\hat{w}_t^G = \hat{w}_t^{G,1} \cup \dots \cup \hat{w}_t^{G,M}$$

Outcome:

- ▶ Fair global model used as initialization for next local training round.
- ▶ Improves performance fairness across hospitals while maintaining global accuracy.

Experiment: Datasets

Dataset	Clients	Image Size	Train/Val/Test Split
ProstateMRI [18]	6	384×384	6:2:2
RIF [29]	4	384×384	4:1 (training split), testing pre-defined

Metrics:

- ▶ Dice coefficient for segmentation accuracy
- ▶ Standard deviation across clients for performance fairness

Experiment: Baselines

Method	Description
Solo	Clients train independently, no communication
FedAvg [21]	Standard FL with global averaging
q-FedAvg [16]	Client re-weighting via empirical loss
CFFL [20]	Re-allocates server model using validation contribution
CGSV [31]	Cosine gradient Shapley value for contribution estimation
Ditto [14]	Personalized fair FL with client-specific optimization
FedCE [8]	Client re-weighting using gradients and validation loss (Sum./Multi.)

Results: Dice Scores (%) for ProstateMRI and RIF

Method	Sites	Dice Scores (%)	Avg.	Std.
Solo	ProstateMRI: Site1-6	84.41, 85.02, 93.70, 91.06, 89.33, 85.43	88.16	3.46
	RIF: Site1-4	82.01, 74.57, 91.26, 92.56	85.10	7.31
FedAvg [21]	ProstateMRI: Site1-6	85.84, 92.27, 93.58, 87.89, 92.47, 80.66	88.78	4.54
	RIF: Site1-4	82.70, 72.68, 91.19, 91.93	84.63	7.79
q-FedAvg [16]	ProstateMRI: Site1-6	85.84, 94.93, 90.22, 88.53, 90.43, 83.32	88.88	3.67
	RIF: Site1-4	77.83, 80.48, 91.72, 91.62	85.41	6.32
CFFL [20]	ProstateMRI: Site1-6	82.84, 94.49, 89.83, 86.89, 90.83, 82.59	87.91	4.29
	RIF: Site1-4	81.49, 74.97, 91.33, 89.53	84.33	6.55
CGSV [31]	ProstateMRI: Site1-6	82.36, 91.02, 90.14, 87.25, 91.06, 82.71	87.42	3.68
	RIF: Site1-4	78.79, 77.30, 89.77, 91.78	84.41	6.42
Ditto [14]	ProstateMRI: Site1-6	88.15, 93.49, 92.72, 89.99, 92.62, 83.64	90.10	3.42
	RIF: Site1-4	82.11, 78.91, 90.90, 91.95	85.97	5.58
FedCE (Sum.) [8]	ProstateMRI: Site1-6	88.13, 94.02, 91.14, 90.63, 92.27, 87.93	90.69	2.15
	RIF: Site1-4	87.02, 76.60, 90.92, 90.28	86.21	5.73
FedCE (Multi.) [8]	ProstateMRI: Site1-6	89.85, 93.49, 92.21, 90.67, 93.66, 89.53	91.57	1.65
	RIF: Site1-4	86.00, 77.42, 90.69, 89.93	86.01	5.26
Fed-LWR	ProstateMRI: Site1-6	91.92, 94.73, 93.22, 93.34, 94.06, 92.21	93.25	0.97
	RIF: Site1-4	85.89, 79.43, 92.42, 90.96	87.17	5.08

Analytical Studies: Ablation Study

Purpose: Examine the design of Fed-LWR (feature difference measurement and layer-wise aggregation).

Variants:

- ▶ **Fed-LWR-v1:** Uses **cosine similarity** instead of CKA to measure feature differences.
- ▶ **Fed-LWR-v2:** Uses only the **last layer's CKA similarity** of U-Net encoder for aggregation.

Insights:

- ▶ Cosine similarity performs worse than CKA for estimating feature differences.
- ▶ Layer-wise aggregation significantly improves fairness and performance.
- ▶ Fed-LWR achieves the best balance of accuracy and fairness.

Ablation Study: ProstateMRI Dice Scores (%) (2/2)

Method	Site1	Site2	Site3	Site4	Site5	Site6	Avg.	Std.
FedAvg [21]	85.84	92.27	93.58	87.89	92.47	80.66	88.78	4.54
Fed-LWR-v1	85.76	94.29	94.36	88.51	92.91	86.33	90.36	3.62
Fed-LWR-v2	89.27	94.86	92.99	90.50	93.24	89.32	91.70	2.12
Fed-LWR	91.92	94.73	93.22	93.34	94.06	92.21	93.25	0.97

Table 3: Dice scores (%) on the testing set of ProstateMRI for different Fed-LWR variants. Avg. and Std. represent the average and standard deviation across clients. Best results are highlighted in bold.

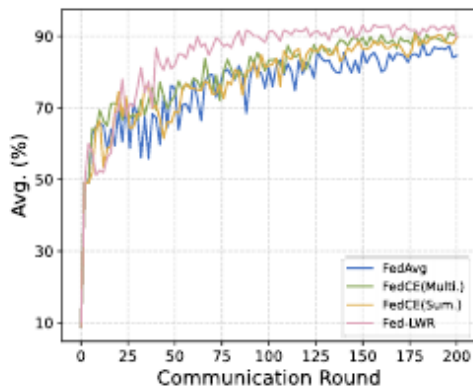
Analytical Studies: Convergence Analysis (1/2)

Objective: Compare convergence of Fed-LWR, FedAvg, and FedCE.

Observations:

- ▶ Fed-LWR converges faster; stable after ~ 90 communication rounds.
- ▶ Fairness mechanism does not slow down convergence.

Visualization: Testing performance vs communication rounds



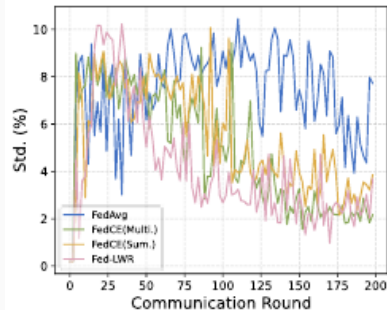
Analytical Studies: Convergence Analysis (2/2)

Observation of Variance:

- ▶ Fed-LWR shows larger standard deviation between rounds 20–50 due to rapid convergence.
- ▶ This variance reflects the adaptive fairness adjustments rather than instability.

Conclusion:

- ▶ Fed-LWR achieves both **fast convergence** and **improved fairness** across clients.
- ▶ Layer-wise re-weighting effectively balances accuracy and fairness without sacrificing global performance.



Limitations of Fed-LWR

Privacy Concerns:

- ▶ CKA requires sharing layer activations → potential leakage of sensitive features.

Rigid Layer Reweighting:

- ▶ All layers are reweighted equally, ignoring task relevance or critical layers.

Limited Task Domain Generalization:

- ▶ Originally evaluated only on segmentation; may not handle classification, detection, or heterogeneous label distributions.

Fixed Similarity Metric:

- ▶ Uses CKA only; may not capture all non-linear or distributional differences effectively.

Hierarchical Fed-LWR: Concept & Motivation

Multi-level FL architecture:

- ▶ Hospitals → Regional aggregators → Global server.
- ▶ Regional aggregation reduces communication load and aligns features before global aggregation.

Motivation:

- ▶ Apply layer-wise reweighting locally and regionally → improve fairness while maintaining scalability.
- ▶ Reduce raw feature exposure at global level → enhanced privacy.

Key novelty: Integrates Fed-LWR's feature-shift aware weighting with hierarchical aggregation for secure, scalable, and fair FL.

Hier-Fed-LWR: Privacy-Preserving FL

- ▶ **Hierarchical Aggregation:** Hospitals → Regional aggregators → Global server.
- ▶ **Reduced Feature Exposure:** Layer-wise reweighting done locally and regionally; raw activations never leave hospitals.
- ▶ **Secure Computation:** DP or SMPC can be applied at group level to protect divergence scalars.
- ▶ **Privacy Advantage:** Root/global server sees only aggregated statistics → difficult to invert to individual hospital features.

Hier-Fed-LWR: Privacy Insights

- ▶ **Local Privacy:** No raw features leave hospital; only protected summaries sent.
- ▶ **Regional Privacy:** Aggregators receive group-level statistics; individual contributions hidden.
- ▶ **Reduced Attack Surface:** Hierarchical design limits inversion or membership inference attacks.
- ▶ **Scalable Privacy:** Supports large federations without compromising layer-wise fairness signals.

 Yunlu Yan, Lei Zhu, Yuexiang Li, Xinxing Xu, Rick Siow Mong Goh, Yong Liu, Salman Khan, and Chun-Mei Feng.

A new perspective to boost performance fairness for medical federated learning.

In *Proceedings of the Medical Image Computing and Computer-Assisted Intervention (MICCAI) 2024*, volume LNCS 15010, pages 13–23. Springer Nature Switzerland, 2024.
<https://arxiv.org/abs/2410.19765>.